

THILANI DILMANI SENARATH

Data Analyst

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thilani-dilmani-portfolio-exb4.vercel.app

PROFESSIONAL SUMMARY

Final-year BSc (Hons) Data Science undergraduate (GPA 3.41/4.00) with practical experience in SQL, Python, Power BI, and Tableau gained through academic projects. Skilled in data cleaning, exploratory data analysis, statistical analysis, and dashboard reporting and visualization, with a strong foundation in turning raw datasets into clear, actionable insights. A detail-oriented problem-solver and effective communicator, seeking a Data Analyst internship to apply and grow analytical skills in a real-world business setting.

TECHNICAL SKILLS

- **Data Analysis & Visualization:** SQL, Python, Power BI, Tableau, Excel, Data Cleaning, EDA, Statistical Analysis, Business Intelligence
- **Machine Learning & Modeling:** Scikit-learn, Predictive Analytics, Feature Engineering, Random Forest, NLP, Machine Learning, Hugging Face Transformers, XLM-RoBERTa
- **Tools & Platforms:** Git/GitHub, MLflow, DagsHub, Apache Airflow, REST API, Docker, Supabase, FastAPI, DVC, PySpark, MongoDB Atlas, Parquet, Gemini API, YouTube Data API v3, Blockchain
- **Web & App Development:** React, Flutter, Firebase, HTML, CSS

PROJECTS

ScamShield: An Integrated AI and Blockchain Solution for Multi-Category Scam Detection and Immutable Scam Reporting (Final Year Project , Ongoing)

- Leading data collection, preprocessing, and feature engineering for a multi-source dataset used to detect phishing URLs, SMS spam, and fake accounts.
- Applying machine learning and NLP techniques to classify and score scam risk, with confidence-based verification, reporting dashboards, and blockchain-based storage.

Technology: Python, Scikit-learn, NLP, Flutter, Supabase, Git/GitHub, Blockchain

Public Pulse: Grounded LLM System for Sri Lankan Public Discourse Analysis (Group Project — Ongoing)

- Designed a multilingual pipeline fine-tuning XLM-RoBERTa to classify Sri Lankan YouTube comments (Sinhala, Singlish, English) into social-issue categories.
- Built a grounded LLM layer (Gemini/Groq) generating insights strictly from verified comment evidence, reducing hallucination risk.
- Delivered an end-to-end architecture from YouTube Data API ingestion to a public dashboard with summaries and trend analysis.

Technology: Python, XLM-RoBERTa, Hugging Face Transformers, Gemini API, YouTube Data API v3, NLP

Telco Customer Churn Analysis Project (Group Project)

- Built a complete data analytics pipeline from raw data ingestion to model deployment on a real-world telecom customer dataset.
- Implemented experiment tracking, workflow orchestration, and a REST API to deliver real-time, data-driven predictions for stakeholder reporting.
- Evaluated a Random Forest model achieving 0.8624 ROC-AUC as the best-performing model; contributed to training three models and tracking results with MLflow.

Technology: Python, Scikit-learn, MLflow, DagsHub, Apache Airflow, FastAPI, DVC, Docker

E-Commerce Big Data Retail Analytics Pipeline (Group Project)

- Built a distributed PySpark pipeline using Medallion architecture to clean 541,909 UK e-commerce records.
- Reduced raw data to 392,690 validated records via rule-based checks and SHA-256 duplicate detection.
- Delivered RFM segmentation for 4,338 customers and 3,665 products using Spark SQL and MongoDB Atlas.

Technology: PySpark, MongoDB Atlas, Parquet, Python, Google Colab

Boston Housing Market Analytics Project (Individual Project)

- Analyzed a multi-variable housing dataset, performing end-to-end data cleaning, transformation, and EDA to identify key drivers of property price variation.
- Designed and built interactive Power BI dashboards with KPI tracking to communicate trends clearly to non-technical stakeholders.

Technology: Power BI, Python, EDA

EDUCATION

BSc (Hons) in Data Science — SLTC Research University, Padukka, Sri Lanka

October 2023 – Present | GPA 3.41/4.00

Relevant coursework: Machine Learning, Deep Learning, Tensors & Graphs, Databases, Statistical Analysis

G.C.E. Advanced Level (A/L) — Co/Yasodara Vidyalaya

December 2022 | Physical Science Stream: Combined Mathematics, Physics, Chemistry

CERTIFICATIONS

- **SQL Basic Certification — HackerRank**
- **SQL Intermediate Certification — HackerRank**
- **SQL Advanced Certification — HackerRank**
- **Introduction to Tableau Certification**
- **Beginners Power BI Certification**

SOFT SKILLS & ACHIEVEMENTS

Soft Skills: Problem-Solving, Critical Thinking, Presentation Skills, Teamwork, Communication, adaptability

- **IEEEExtreme 19.0 Programming Competition** — Participant
- **BIOFUSION Hackathon**, University of Sri Jayewardenepura — Participant
- **InspiHER {Tech} v4.0 HackerRank Competition** — Finalist, Top Team

VOLUNTEERING

- **InspiHER Tech v3.0** — Logistics Team Member
- **IEEE Student Branch** — Member
- **Master Designer v3.0** — Content & Caption Team Member
- **DECODE XTREME v2.0** — Assistant Secretary
- **ITIA Career v1.0** — Content & Caption Deputy Head

REFERENCES

Available Upon Request